

A reproducible Climate Risk & Vulnerability Assessment for the seven EW4All pilot countries

Methods note · real INFORM Risk + World Bank Open Data · github.com/PrOspektor/climate-risk-early-warning

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1. Purpose

This note documents the method behind an open, reproducible Climate Risk & Vulnerability Assessment (CRVA) for the seven Green Climate Fund Early Warnings for All (EW4All) pilot countries — Antigua and Barbuda, Cambodia, Chad, Ecuador, Ethiopia, Fiji and Somalia. It answers a prioritisation question: where do high hazard & exposure, high human vulnerability, and weak basic systems (power, water, mobile reach) coincide, so an early-warning programme should act first?

2. Data & provenance

- **Hazard:** Hazard & Exposure (and full component decomposition): JRC INFORM Risk 2023 (API WorkflowId 464), 0–10.
- **Vulnerability:** Vulnerability: World Bank Open Data — GDP per capita (inverted), undernourishment, under-5 mortality, rural population share.
- **Readiness:** Readiness gap: World Bank Open Data — access to electricity, basic drinking water, mobile subscriptions per 100 (all inverted; mobile capped at 100 = saturated reach, a proxy for warning reach).

All values are official open data, fetched live (accessed 2026-07-06; World Bank last-updated 2026-07-01). Every indicator code, year and URL is listed in `data/sources.md`, and `src/fetch_data.py` regenerates the dataset from the World Bank and INFORM APIs. Missing values (Antigua's undernourishment and mobile figures) are left blank and excluded from that country's dimension average — never imputed.

3. Method

Each indicator's polarity is set so higher = worse, then min-max normalised to 0–1 across the seven countries (a relative, prioritisation index). A dimension score is the mean of its available indicators; the composite is the arithmetic mean of the three dimensions $\times 100$. An arithmetic mean is used rather than a geometric mean so a country that is best-in-class on one dimension is not forced to an artificial zero. The approach follows the OECD/JRC Handbook on Constructing Composite Indicators.

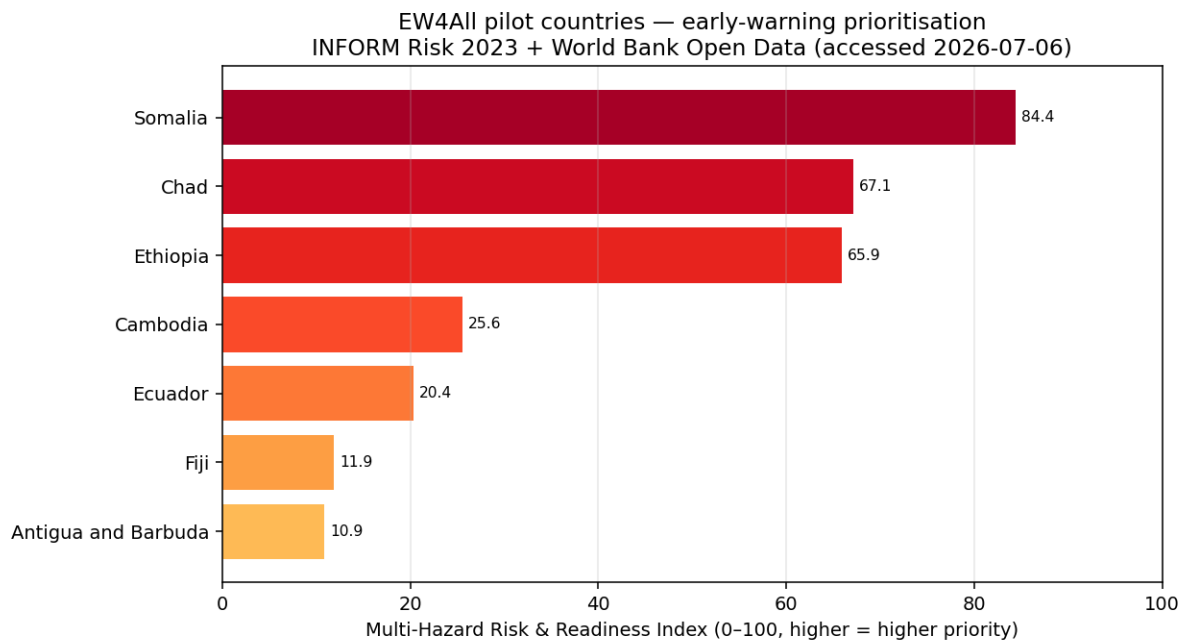


Fig. 1 — Composite index and priority ranking (higher = higher priority).

4. Independent validation

The composite is cross-checked against the JRC INFORM overall risk score, which is not used in its construction. The two — built from different inputs — agree at a Spearman rank correlation of $\rho \approx 0.99$, a genuine external check rather than a circular one.

5. Uncertainty, rank-stability & sensitivity

A 20,000-run Monte-Carlo simulation draws the three dimension weights from a Dirichlet distribution and perturbs every normalised indicator with Gaussian noise. Because the score gap from rank 3 to rank 4 (~40 pts) is ~7× the simulation spread (~6 pts), top-3 membership is trivially stable; the decision-relevant structure is the within-block ordering, shown by the full rank-probability surface (Fig. 2): Somalia is #1 in ~98% of runs, while Chad and Ethiopia trade ranks 2–3 and Antigua/Fiji are a ~50/50 tie for 6th/7th.

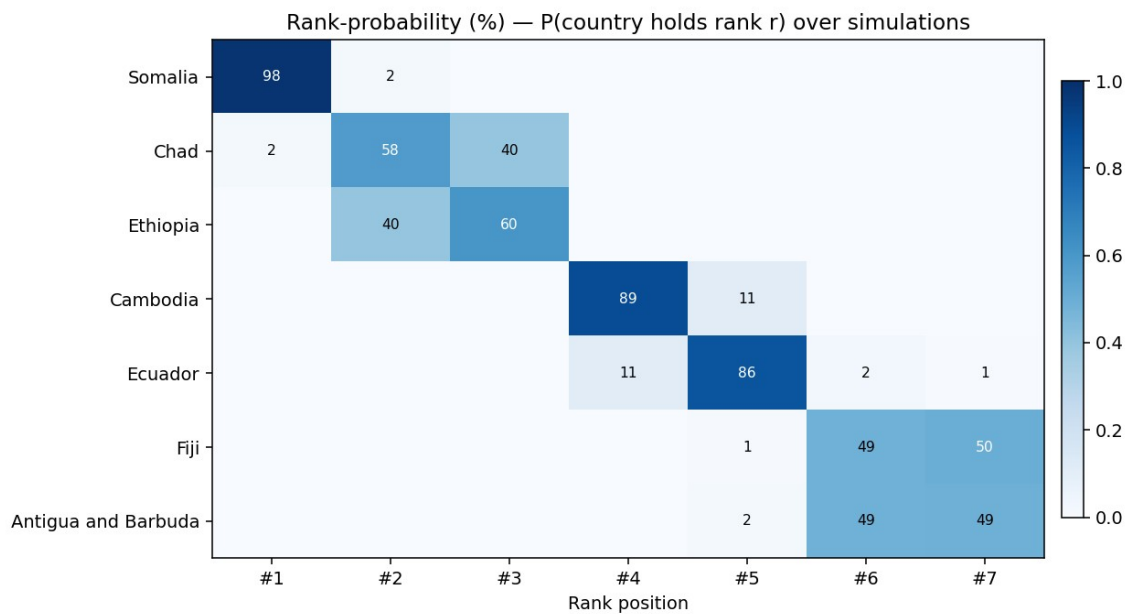


Fig. 2 — Rank-probability: $P(\text{country holds rank } r)$ across simulations.

The interval on each score is a sensitivity band, not a statistical confidence interval — the indicators are census-style values with no sampling error. A variance decomposition attributes the band mostly to the weighting choice (e.g. 91% for Chad), largest where a country's dimension profile is uneven (Fig. 3). Structural robustness is confirmed by Kendall's $\tau = 0.90\text{--}1.00$ across geometric vs arithmetic aggregation and z-score vs min-max normalisation.

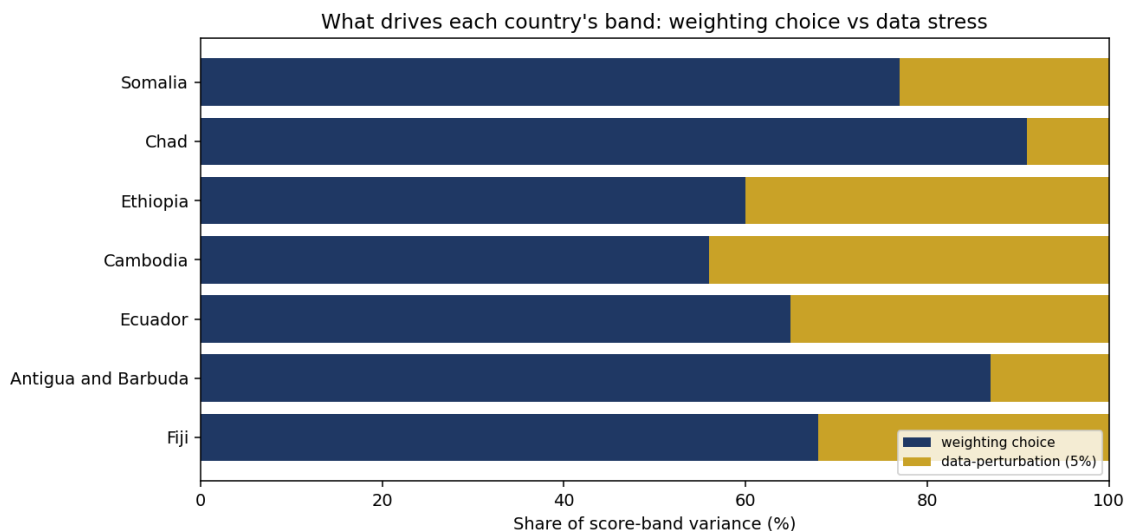


Fig. 3 — Variance in each score band: weighting choice vs data-perturbation.

6. From assessment to early warning

The CRVA identifies where to prioritise; a companion module (`src/rainfall_ew.py`) demonstrates the observation→warning step using real ERA5 reanalysis rainfall (Open-Meteo). For each pilot capital it fires flood alerts when rolling 3-day accumulation exceeds the site's own p95/p99 and flags drought from consecutive dry-day runs; self-calibrating percentile thresholds let the same logic transfer from the Sahel to the Pacific unchanged (EW4All Pillars 2–3).

7. Limitations & honest scope

- Scores are relative across the seven pilot countries, not absolute global risk values.
- The composite uses INFORM for hazard and World Bank for vulnerability/readiness; a fuller build could add sub-national resolution and observed loss data (EM-DAT).
- The rainfall module is a capital-point demonstrator, not an operational forecast system.

8. Reproducibility

Python (NumPy/pandas/matplotlib), no proprietary dependencies. `fetch_data.py` → `build_index.py` → `spatial_map.py` → `uncertainty.py` → `rainfall_ew.py` regenerate every figure and table; a one-click Colab notebook is provided. Code is MIT-licensed and archivable to a citable DOI (see [RELEASING.md](#)).